Framework for planning the training sessions in triathlon

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Agenda

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- Origins of triathlon
- Proposed framework:
  - data collection
  - planning the training sessions
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- Experiments and results
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Introduction

- Sophisticated AI methods assist athletes in planning sports training sessions (e.g., in individual sports disciplines, like: running, cycling etc.)
- Creating efficient training plans demands:
  - studying various new sport methods, and
  - knowledge discovering from data about completed sports activities.
- The first demand is coping in domain of sport theory
- The second demand is solving in domain of computer science, where the following AI methods have been applied recently:
  - Artificial Neural Networks (ANN),
  - Evolutionary Algorithms (EA),
  - Swarm Intelligence based algorithms (SI).

Motivation

- Solving the problem of planning training session in triathlon
- Triathlon is multi-sports discipline consisting of:
  - swimming,
  - cycling,
  - running.
- Triathlon is a very complex problem from the training point of view due to:
  - usually, sports trainers prefer swimming training sessions before running,
  - cycling training is more important than the running.
- Developing the framework of SI-based algorithms capable of:
  - monitoring realization of training activities by triathlon athletes,
  - collecting data obtained in this process,
  - planning training plans based on these data.

Triathlon

- Triathlon is the hardest one-day endurance sport
- Ironman triathlon came into being on February 18, 1978
- It combines three marathon distances in swimming (3.8 km), cycling (180 km), and running (42.195 km)
- Ironman was popularized by the US television network ABC
- Exciting moment in live broadcast was, when Julie Moss in 1982 crossing crawling across the finish line and won the hearts of millions of viewers
- Each year, final race in Ironman returns symbolically to its roots on Big Island on Hawaii





The purpose

- The framework is dedicated to planning the training sessions in triathlon
- The results of performing the training plan must be a simultaneous development of athlete's psycho-physical performances
- For each day, there is a combination of training plans for three different disciplines
- Each specific discipline has its own goal that needs to be followed during planning the training sessions
- The framework consists of three processes, as follows:
  - data collection,
  - planning the training sessions,
  - evaluation.

Data collection options

- Typically, sports activities are monitored with mobile devices (e.g., sports watches, smart phones) worn by athlete during training session
- Data in swimming are collected manually, because the majority of classical mobile devices do not support this discipline
- These data are collected and saved into databases
- Data includes measured values of training load indicators, like duration, distance, average speed, burned calories etc.
- In our study, data originate from two sources:
  - data referring to swimming were obtained manually,
  - data in cycling and running were obtained automatically.

Problem description

- Planning the training sessions is defined as an optimization problem
- Intention: generate training plans for all three disciplines in training cycle
- Input: training cycle of duration d, database of performed sports training sessions
- Criteria: intensity of training session expressed as

$$TRIMP = TD \cdot HR,$$

  where $TRIMP$ is training load indicator, TD training duration, and HR the average heart rate.
- Task: maximize the value of $TRIMP$
- Output: generated training plan for the training cycle

Particle Swarm Optimization (PSO)

- PSO was proposed by Eberhart and Kennedy in 1995
- It is a population-based SI-based algorithm that works on real-coded particles:

$$\mathbf{x}_i^{(t)} = (x_{i,1}^{(t)}, \ldots, x_{i,d}^{(t)}), \quad \text{for } i = 1, \ldots, NP,$$

where NP denotes the population size.

- Moving the particles in PSO is expressed mathematically as:

$$\begin{aligned} \mathbf{v}_i^{(t+1)} &= w \cdot \mathbf{v}_i^{(t)} + c_1 \cdot rand_1 \cdot (\mathbf{x}_{best} - \mathbf{x}_i^{(t)}) + c_2 \cdot rand_2 \cdot (\mathbf{p}_i^{(t)} - x_i^{(t)}), \\ \mathbf{x}_i^{(t+1)} &= \mathbf{x}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \end{aligned}$$

where

$\mathbf{x}_i^{(t)}$	– position of particle	$\mathbf{v}_i^{(t)}$	– velocity of particle
$\mathbf{p}_i^{(t)}$	– personal best solution	$\mathbf{g}^{(t)}$	– global best solution
$\mathbf{c}_1, \mathbf{c}_2$	– cognitive and social coefficient	$\mathbf{w}$	– inertia weight
$rand_1, rand_2$	– random numbers		

Pseudo-code of PSO

Algorithm 1 Pseudo-code of the basic PSO algorithm

```
1: init_particles;
2: while termination_condition_not_meet do
3:     for i = 1 to Np do
4:         f_i = evaluate_the_new_solution(x_i);
5:         if f_i ≤ pBest_i then
6:             p_i = x_i; // save the local best solution
7:             pBest_i = f_i;
8:         end if
9:         if f_i ≤ f_min then
10:            x_best = x_i; // save the global best solution
11:            f_min = f_i;
12:        end if
13:        x_i = generate_new_solution(x_i);
14:    end for
15: end while
```

Representation

- Representation of particles was extended for 2 elements:

$$\mathbf{x}_i^{(t)} = (x_{i,1}^{(t)}, \ldots, x_{i,d}^{(t)}, x_{i,d+1}^{(t)}, x_{i,d+2}^{(t)}), \quad \text{for } i = 1, \ldots, NP,$$

where

$x_{i,j}^{(t)} \in [0,1]$ – encodes index of training session ID_k in an archive A_k,
$x_{i,d+1}^{(t)}$ – the first cut point,
$x_{i,d+2}^{(t)}$ – the second cut point.

- Index of training session is decoded as:

$$ID_k = \lfloor x_{i,j}^{(t)} \cdot |A_k| \rfloor + 1,$$

where $k \in \mathcal{D}$ and $\mathcal{D} = \{swim, cycle, run\}$.

Genotype-phenotype mapping

- Both cutting points need satisfy the following objectives:

$$\begin{aligned} 3 \cdot q_{\text{min}} &\leq d, \\ q_1 &\geq q_{\text{min}}, \\ q_2 - q_1 &\geq q_{\text{min}}, \\ d - q_2 &\geq q_{\text{min}}, \end{aligned}$$

  where q_{min} denotes the minimum training days.
- The cut points are expressed according to the equations:

$$\begin{aligned} q_1 &= q_{\text{min}} + \lfloor x_{i,d} \cdot |I_1| \rfloor + 1, \text{ and} \\ q_2 &= q_1 + \lfloor x_{i,d+1} \cdot (d - |I_2|) \rfloor + 1, \end{aligned}$$

  where $|I_1| = d - 3 \cdot q_{\text{min}}$ and $|I_1| = d - q_1 - 2 \cdot q_{\text{min}}$.
- Thus, it holds:

$$\begin{aligned} q_1 &\in [q_{\text{min}}, d - 2 \cdot q_{\text{min}}], \text{and} \\ q_2 &\in [2 \cdot q_{\text{min}}, d - q_{\text{min}}]. \end{aligned}$$

Fitness function evaluation

- Fitness function is evaluated as weighted sum of *TRIMP* training load indicators:

$$f(\mathbf{x}_i) = \max \sum_{k \in \mathcal{D}} w_k \cdot TRIMP_k(\mathbf{x}_i),$$

where

w_k – the importance of the sports discipline in the training,
$TRIMP_k(\mathbf{x}_i)$ – proposed training load indicator in the particular sports discipline.

- The proposed training load indicator is calculated as:

$$TRIMP_k(\mathbf{x}_i) = \sum_{day=1}^{d} TRIMP(ID_k), \quad \text{for } k \in \mathcal{D},$$

where

day – training days in the period of duration d,
$TRIMP(ID_k)$ – the *TRIMP* load indicator in the particular sports discipline.

Evaluation of training plan

- The real evaluation of training plan is demonstrated on competitions
- The improvement of the athletes can only be performed after completion of the training period
- In this evaluation, data collected during the performed training sessions are compared with the expected results of sports trainers
- In our study, the generated training plans were informally given to the real sports trainers for commenting

PSO parameter setting

Parameter	Symbol	Value
Population size	NP	100
Dimension of the problem	d	50
Inertia weight	w	0.7298
Attraction strength 1	c_1	1.496
Attraction strength 2	c_2	1.496
Minimum training days	$q_{\min}$	12
Weight of swimming	w_{swim}	1.0
Weight of cycling	w_{cycle}	0.5
Weight of running	w_{run}	0.75

- The number of independent runs was 30
- Only one solution was taken into analysis in the study

Archive of basic sports activities

Nr.	Swimming			Cycling			Running		
	ID	HR	TD	ID	HR	TD	ID	HR	TD
1	SW-1	142	65	CY-1	125	70	RU-1	125	38
2	SW-2	141	91	CY-2	130	81	RU-2	121	40
3	SW-3	141	74	CY-3	138	120	RU-3	141	71
4	SW-4	142	65	CY-4	145	124	RU-4	138	87
5	SW-5	156	64	CY-5	143	110	RU-5	145	84
6	SW-6	141	55	CY-6	160	90	RU-6	150	65
7	SW-7	158	47	CY-7	165	119	RU-7	162	82
8	SW-8	125	38	CY-8	167	134	RU-8	167	61
9	SW-9	128	36	CY-9	172	60	RU-9	173	52
10	SW-10	128	26	CY-10	173	94	RU-10	175	40

- The archive consists of 10 basic sports activities that were collected manually

Swimming

Nr.	ID	*TRIMP*/session	Repetition	*TRIMP*/period
1	SW-3	10,434	3	31,302
2	SW-4	9,230	3	27,690
3	SW-5	9,984	8	79,872
4	SW-6	7,755	3	23,265
5	SW-7	7,426	1	7,426
6	SW-8	4,750	1	4,750
7	SW-9	4,608	1	4,608
$\sum$	n/a	54,187	20	1,083,740

Cycling

Nr.	ID	*TRIMP*/session	Repetition	*TRIMP*/period
1	CY-3	16,560	1	16,560
2	CY-4	17,980	3	53,940
3	CY-5	15,730	2	31,460
4	CY-6	14,400	1	14,400
5	CY-7	19,635	1	19,635
6	CY-8	22,378	3	67,134
7	CY-10	16,262	1	16,262
$\sum$	n/a	122,945	12	1,475,340

Running

Nr.	ID	*TRIMP*/session	Repetition	*TRIMP*/period
1	RU-3	10,011	1	10,011
2	RU-4	12,006	1	12,006
3	RU-5	12,180	5	60,900
4	RU-7	13,284	10	132,840
5	RU-8	10,187	1	10,187
$\sum$	n/a	57,668	18	1,038,024

Analysis of the proposed training plan

- Problem: How to balance the *TRIMP* load indicator values between all three sports disciplines
- Results of the analysis: The swimming sessions are less intensive than running and cycling, but cycling session are more intensive than running
- Solution: Introducing weights

Discipline	*TRIMP*	Sess.	*TRIMP*/Sess.	Weights	Ratio
Swimming	1,083,740	20	54,187	1.00	0.30
Cycling	1,475,340	12	122,945	0.50	0.41
Running	1,038,024	18	57,668	0.75	0.29
Summary	3,597,104	50	234,800	2.25	1.00

- Finding: Not perfectly balanced! While swimming and running are balanced, cycling is overestimated.

Conclusions

- The paper proposes the new approach of planning the training sessions in triathlon
- The main problem here is to balance the *TRIMP* load indicator between three disciplines
- The proposed framework introduces weights in fitness function evaluation, which regulate too big impact of the particular discipline
- The results showed that the generated training plans are suitable for using in practice
- In the future, the real archive of performed training activities should be collected
- The other nature-inspired algorithms would be included into the framework